

# Introduction

## Microprocessors and Assembly Language

### **Course Teacher:**

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**Course ID:** CSE - 4503

**Course Title:** Microprocessors and Assembly Language  
Department of Computer Science and Engineering (CSE),  
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# Topics to be Covered in this Course !!

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- ▶ Microprocessor and Assembly Language:
- ▶ Microprocessors and Microcomputers
- ▶ Evaluation of Microprocessors Applications
- ▶ Intel 8086 Microprocessor: internal architecture, register structure, programming model, addressing modes, instruction set
- ▶ Assembly language programming
- ▶ Coprocessors.
- ▶ An overview of Intel 80186, 80286, 80386, 80486 and Pentium microprocessors
- ▶ RISC and CISC processors.

# Recommended Texts

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- ▶ *Microprocessor, architecture, programming & application with the 8085*, **Author:** Gaonkar
- ▶ *Microprocessors and Interfacing: Programming and Hardware*, **Author:** Douglas V. Hall
- ▶ *Assembly Language Programming and Organization of the IBM PC*, **Author:** Ytha Yu, Charles Marut
  
- ▶ **Additionally !!** Lecture materials will be provided.

# Does Earlier Knowledge Require ??

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- ▶ You should have the knowledge about –
  - ▶ *Number Systems.*
  - ▶ Basics of “*Digital Logic Design*” course.
  - ▶ Basics of “*Computer Organization and Architecture*” course.
  - ▶ “*Basic Programming*”.

# Lecture References:

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## ▶ **Book:**

- ▶ *Microprocessors and Interfacing: Programming and Hardware,*  
**Author:** Douglas V. Hall (Chapter-2)

## ▶ **Lecture Materials:**

- ▶ *ECP2036: Microprocessor and Interfacing.*

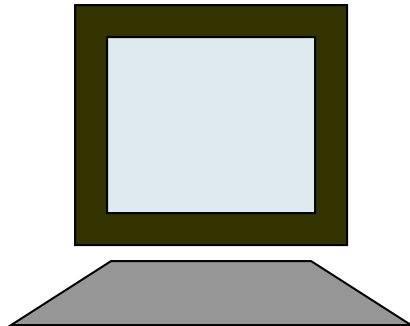
## ▶ **Web Materials:**

- ▶ Wikipedia, the free encyclopedia
- ▶ <http://www.cpu-world.com/Arch/8085.html>
- ▶ [http://www.ehow.com/way\\_5230222\\_8085-microprocessor-tutorial.html](http://www.ehow.com/way_5230222_8085-microprocessor-tutorial.html)

# Concept of Computer

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**Computer**

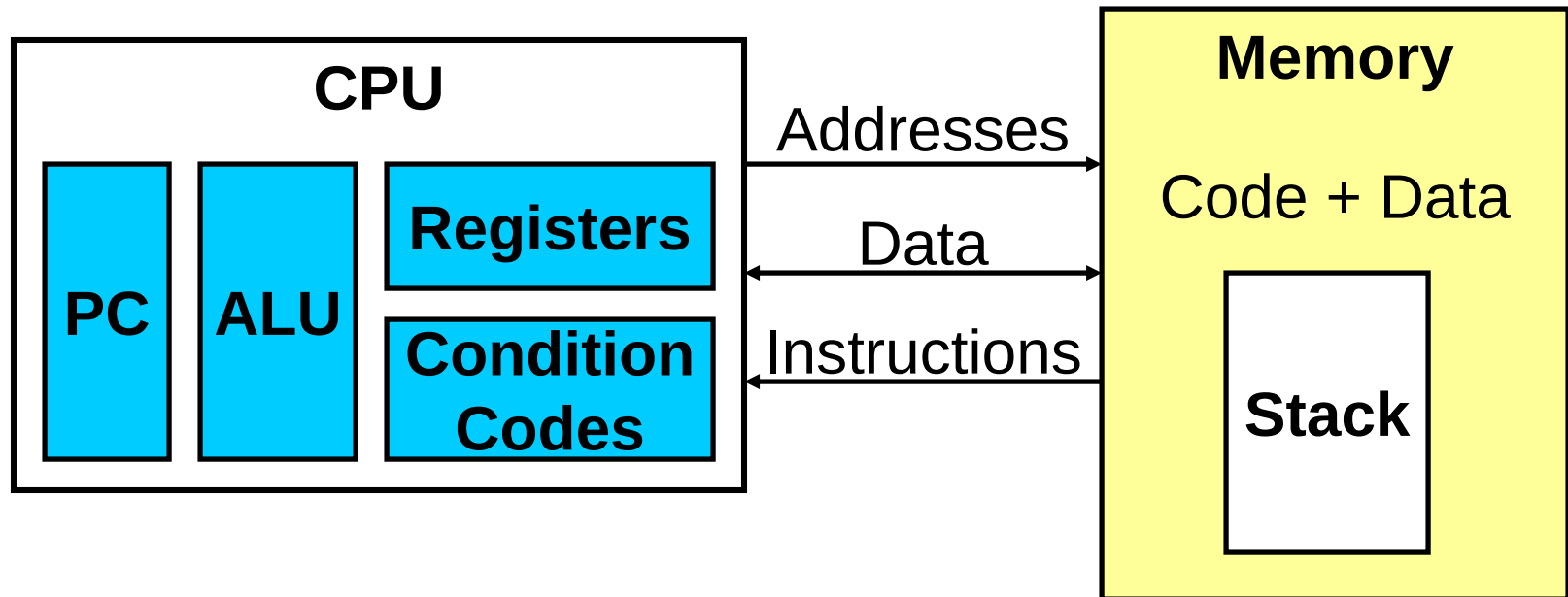


**Data Processing**

**Data Storage**

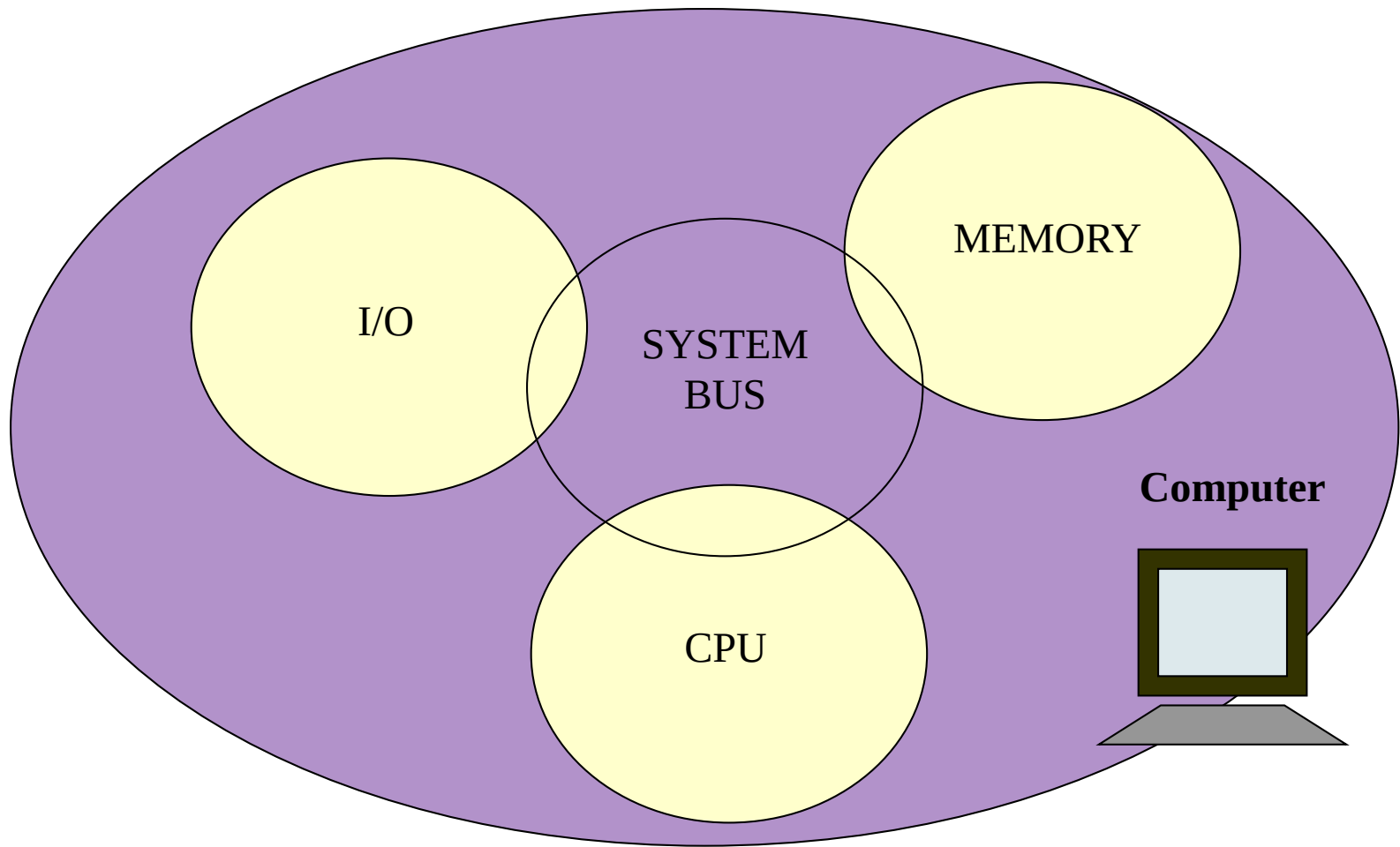
# Major Components of Computer

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# Major Components of Computer

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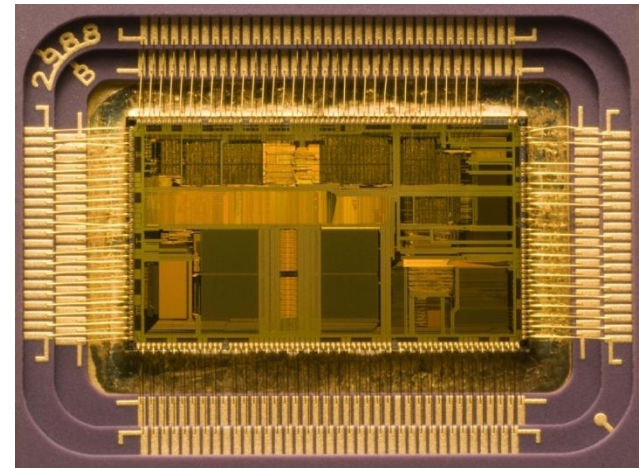
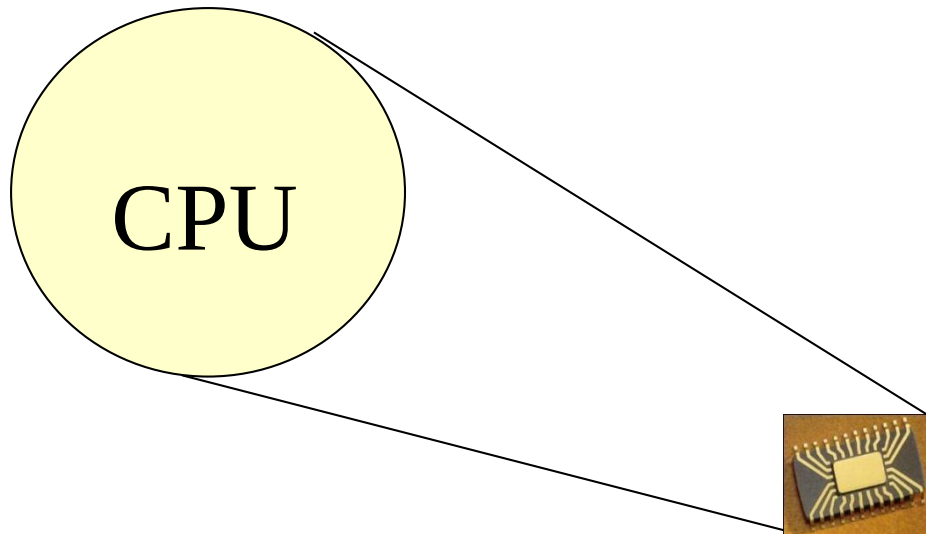




# Concept about Microprocessor

- ▶ A **microprocessor** incorporates most or all of the functions of a central processing unit (CPU) on a single **integrated circuit (IC)**.

Die of an Intel **80486DX2**  
microprocessor (actual size: 12×6.75 mm)  
in its packaging

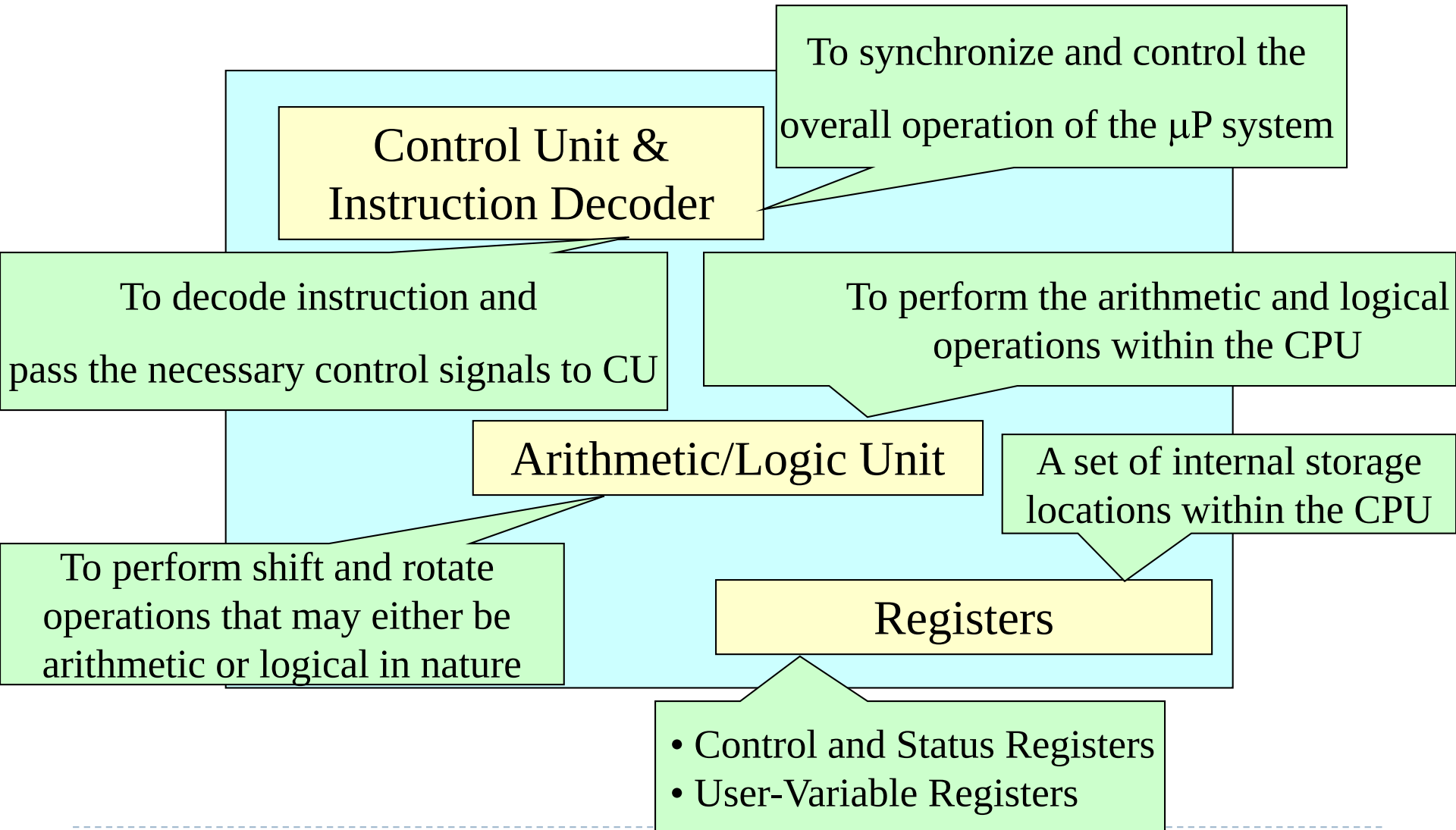


# Central Processing Unit

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- ▶ A **central processing unit (CPU)**, or sometimes just called **Microprocessor ( $\mu$ P)**, is a description of a class of logic machines that can execute **computer programs**.
- This broad definition can easily be applied to many early computers that existed long before the term "CPU" ever came into widespread usage. However, the term itself and its initialism have been in use in the computer industry at least since the early 1960s.
- The form, design and implementation of CPUs have changed dramatically since the earliest examples, but their fundamental operation has remained much the same.

# Central Processing Unit/Microprocessor ( $\mu$ P)



# So .. What is Microprocessor?

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- ▶ A microprocessor (abbreviated as  $\mu\text{P}$  or  $\text{uP}$ ) is a Silicon Chip that contains an electronic central processing unit (CPU). In the world  $\text{uP}$  or CPU used interchangeably, which is made from miniaturized transistors and other circuit elements on a single semiconductor integrated circuit (IC).
- ▶ The integration of the whole CPU onto a single **VLSI Chip** therefore greatly reduced the cost of processing capacity.
- ▶ **Architectures of Microprocessors:**
  - ▶ 8-bit designs
  - ▶ 16-bit designs
  - ▶ 32-bit designs
  - ▶ 64-bit designs
  - ▶ Multi-core designs
  - ▶ RISC (Reduced Instruction Set Computer)
  - ▶ CISC (Complex Instruction Set Computer)
  - ▶ Special-purpose designs: Microcontrollers, Digital Signal Processors (DSP) and Graphics Processing Units (GPU).

# List of Microprocessors

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**1971** - Intel 4004, 1st single chip CPU, 4-bit processor, 45 instructions

**1972** - Intel 4040, enhanced 4004, 60 instructions

**1972** - Intel 8008, 8-bit  $\mu$ P

**1972** - Texas Instrument TMS 1000, 1st single  $\mu$ C, 4-bit

**1974** - Intel 8080, successor to the 8008, used in Altair 8800

**1975** - Motorola 6800, used MOS technology

**1976** - Intel 8085, updated 8080, +5V power supply

**1976** - Zilog Z80, improved 8080

**1976** - TI TMS 9900, 1st 16-bit  $\mu$ P

**1978** - Zilog Z8000, Motorola 68000, 16-bit  $\mu$ P

**1978** - Intel 8086, 16-bit, IBM's choice...

# History of Intel Era

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- 1978: 8086 (16 bit architecture)
- 1980: 8087
  - Floating point coprocessor is added
- 1982: 80286
  - Increases address space to 24 bits
- 1985: 80386:
  - 32 bits Address Space,
  - Virtual Memory & new addressing modes
  - Protected mode (OS support)
- 1989-95: 80486/Pentium/Pro
  - Added a few instructions of base MMX

# History of Intel Era

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- 1997: Pentium II
  - 57 new “MMX” instructions are added,
- 1999: Pentium III:
  - Out of Order, added another 70 Streaming SIMD Ext (SSE)
- 2001: Pentium 4
  - Net burst, another 144 instructions (SSE2)
- 2003: PI4 HT, Trace Cache
- 2005: Centrino, low power
- 2007: Core architecture, Duo
- 2008: Atom, Quad core with HT....
- 2009 --- till now: Multi core (Large chip multiprocessor)

# Assembly Language

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- ▶ **Assembly language:**

- ▶ Assembly language is used for most programming because it is extremely difficult to program a microprocessor in its native, that is hexadecimal machine language.

- ▶ **Assembler:**

- ▶ An assembler is a program that converts software written in symbolic machine language (the source programme) into hexadecimal machine language (object programme).
- ▶ The primary reason to use assembler is because development and modification are always difficult in machine language.



# Example of 8085 Assembly Language

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| <u>Address</u> | <u>Mnemonics</u> | <u>Machine (Hex) Code</u>                                             |
|----------------|------------------|-----------------------------------------------------------------------|
| 202A           | MVI A, 32H       | 3E<br>32 ;Copies 32H into accumulator                                 |
| 202C           | MVI B, 48H       | 06<br>48 ; Copies 48H into B register                                 |
| 202E           | ADD B            | 80 ;Adds B reg. content with Acc and<br>stores the result in Acc.     |
| 202F           | STA [4I FF]      | D3<br>0I ;Stores the Acc (the sum) into the<br>memory location 4I FF. |
| 203I           | HLT              | 76 ; Stops the program                                                |

# Another Example of 8085 Assembly Language

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| <u>Address</u> | <u>Instruction</u> |                                                                        |
|----------------|--------------------|------------------------------------------------------------------------|
| 2020           | MVI    B, 24       | ; Copies 24 into B register                                            |
| 2022           | INR    B           | ; Increment B reg content by 1                                         |
| 2023           | MOV    A, B        | ; Copies B register into Acc.                                          |
| 2025           | SUB    B           | ; Subtracts B reg content from<br>Acc and stores the result in<br>Acc. |
| 2026           | STA    [5F FF]     | ; Stores the Acc content into<br>the memory location 5F FF.            |
| 2028           | HLT                | ; Stops the program                                                    |

# Solve It !!

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```
MVI  A, 24H      // load Reg ACC with 24H
MVI  B , 56H      // load Reg B with 56H
ADD  B            // ACC= ACC+B
OUT  01H          // Display ACC contents on port 01H
HLT              // End the program
```

OUTPUT ??

# Example of 8086 Assembly Language

- Machine Language vs Assembly Language
  - Machine language or object code is the only code a computer can execute but it is nearly impossible for a human to work with
  - E4 27 88 C3 E4 27 00 D8 E6 30 F4 the object code for adding two numbers input from the keyboard
- When programming a microprocessor, programmers often use assembly language
  - This involves 3-5 letter abbreviations for the instruction codes (mnemonics) rather than the binary or hex object codes

| Address | Hex Object Code |    |  |  | Mnemonics |         | Comment                                          |
|---------|-----------------|----|--|--|-----------|---------|--------------------------------------------------|
|         |                 |    |  |  | Op-Code   | Operand |                                                  |
| 0100    | E4              | 27 |  |  | IN        | AL,27H  | Input first number from port 27H and store in AL |
| 0102    | 88              | C3 |  |  | MOV       | BL,AL   | Save a copy of register AL in register BL        |
| 0104    | E4              | 27 |  |  | IN        | AL,27H  | Input second number to AL                        |
| 0106    | 00              | D8 |  |  | ADD       | AL,BL   | Add contents of BL to AL and store the sum in AL |
| 0107    | E6              | 30 |  |  | OUT       | 30H,AL  | Output AL to port 30H                            |
| 0109    | F4              |    |  |  | HLT       |         | Halt the computer                                |

# Thank You !!

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Good Luck with the course 😊